Machine-Level Programming I: Basics

CSE 238/2038/2138: Systems Programming

Instructor:
Fatma CORUT ERGİN

Slides adapted from Bryant & O'Hallaron's slides

Today: Machine Programming I: Basics

- **History of Intel processors and architectures**
- **Assembly Basics: Registers, operands, move**
- **Arithmetic & logical operations**
- **C, assembly, machine code**

Intel x86 Processors

- **Dominate laptop/desktop/server market**
- **Evolutionary design**
  - Backwards compatible up until 8086, introduced in 1978
  - Added more features as time goes on
    - Now 3 volumes, about 5,000 pages of documentation
- **Complex instruction set computer (CISC)**
  - Many different instructions with many different formats
    - But, only small subset encountered with Linux programs
  - Hard to match performance of Reduced Instruction Set Computers (RISC)
  - But, Intel has done just that!
    - In terms of speed. Less so for low power.

Intel x86 Evolution: Milestones

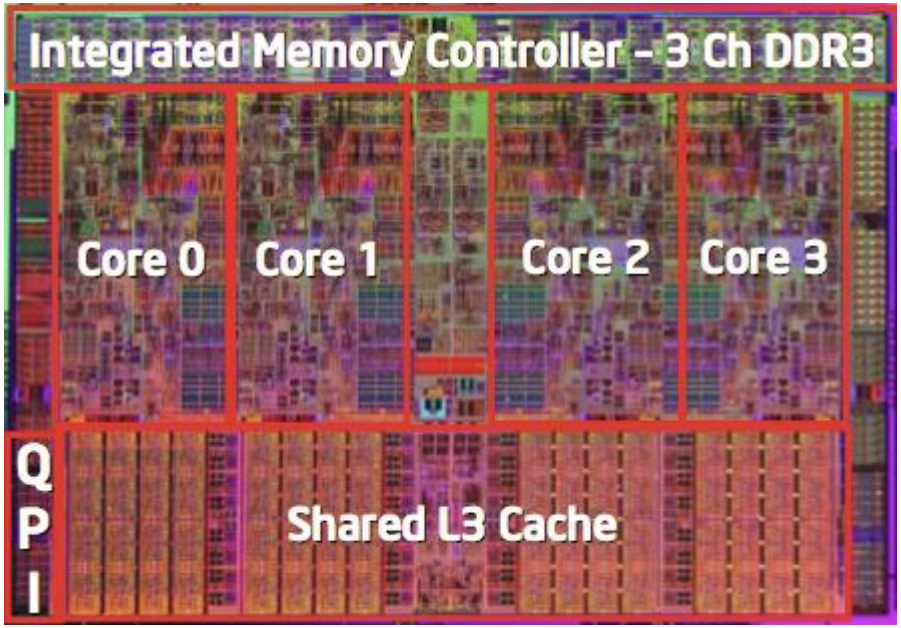
Name	*Date*	*Transistors*	*MHz*
8086	**1978**	**29K**	**5-10**
386	**1985**	**275K**	**16-33**
Pentium 4E	**2004**	**125M**	**2800-3800**
Core 2	**2006**	**291M**	**1060-3500**
Core i7	**2008**	**731M**	**1700-3900**

- First 16-bit Intel processor. Basis for IBM PC & DOS
- 1MB address space
- First 32 bit Intel processor , referred to as IA32
- Added “flat addressing”, capable of running Unix
- First 64-bit Intel x86 processor, referred to as x86-64
- First multi-core Intel processor
- Four cores

Intel x86 Processors, cont.

- **Machine Evolution**

386	1985	0.3M
Pentium	1993	3.1M
Pentium/MMX	1997	4.5M
PentiumPro	1995	6.5M
Pentium III	1999	8.2M
Pentium 4	2001	42M
Core 2 Duo	2006	291M
Core i7	2008	731M
Core i7 Skylake	2015	1.9B



- **Added Features**
  - Instructions to support multimedia operations
  - Instructions to enable more efficient conditional operations
  - Transition from 32 bits to 64 bits
  - More cores

2017 State of the Art

- **Mobile Model: Core i7**
  - 2.6-2.9 GHz
  - 45 W
- **Desktop Model: Core i7**
  - Integrated graphics
  - 2.8-4.0 GHz
  - 35-91 W
- **Server Model: Xeon**
  - Integrated graphics
  - Multi-socket enabled
  - 2-3.7 GHz
  - 25-80 W

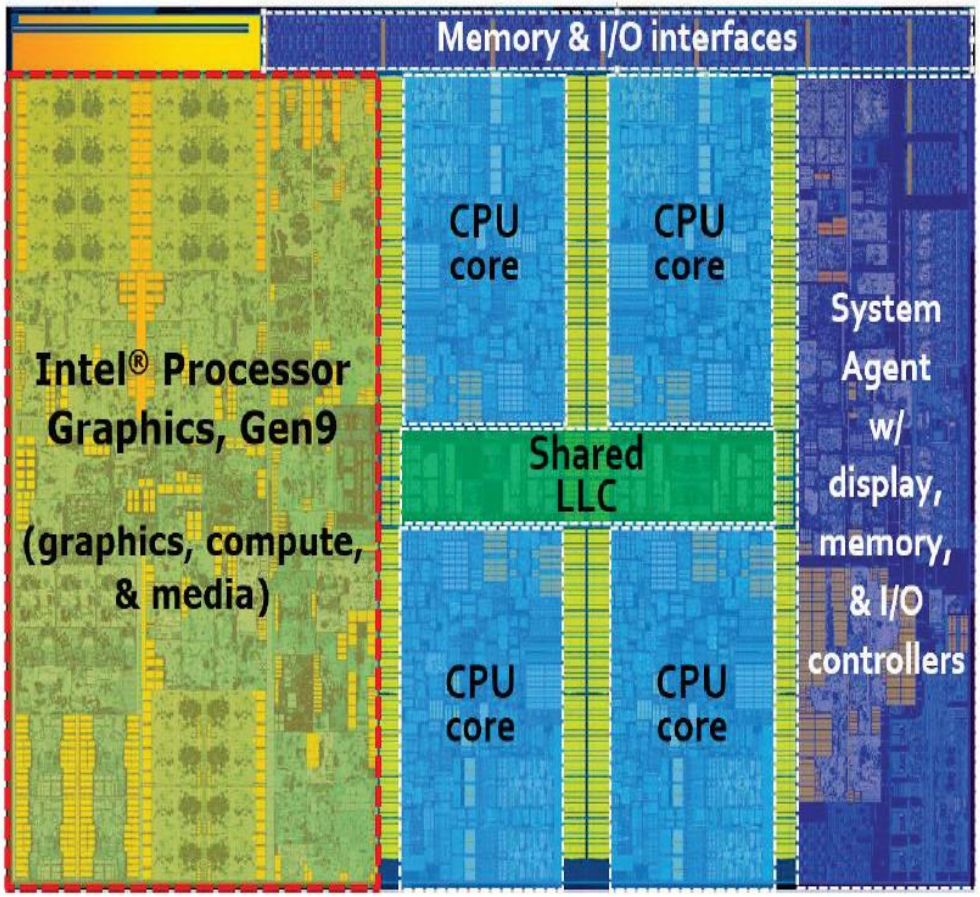


Figure 1: Architecture components layout for an Intel® Core™ i7 processor 6700K for desktop systems. This SoC contains 4 CPU cores, outlined in blue dashed boxes. Outlined in the red dashed box, is an Intel® HD Graphics 530. It is a one-slice instantiation of Intel processor graphics gen9 architecture.

x86 Clones: Advanced Micro Devices (AMD)

- **Historically**
  - AMD has followed just behind Intel
  - A little bit slower, a lot cheaper
- **Then**
  - Recruited top circuit designers from Digital Equipment Corp. and other downward trending companies
  - Built Opteron: tough competitor to Pentium 4
  - Developed x86-64, their own extension to 64 bits
- **Recent Years**
  - Intel got its act together
    - Leads the world in semiconductor technology
  - AMD has fallen behind
    - Relies on external semiconductor manufacturer,

Intel's 64-Bit History

- **2001: Intel Attempts Radical Shift from IA32 to IA64**
  - Totally different architecture (Itanium)
  - Executes IA32 code only as legacy
  - Performance disappointing
- **2003: AMD Steps in with Evolutionary Solution**
  - x86-64 (now called "AMD64")
- **Intel Felt Obligated to Focus on IA64**
  - Hard to admit mistake or that AMD is better
- **2004: Intel Announces EM64T extension to IA32**
  - Extended Memory 64-bit Technology
  - Almost identical to x86-64!
- **All but low-end x86 processors support x86-64**
  - But, lots of code still runs in 32-bit mode

Our Coverage

- **x86-64**
  - The standard
  - `mufe> gcc hello.c`
  - `mufe> gcc –m64 hello.c`

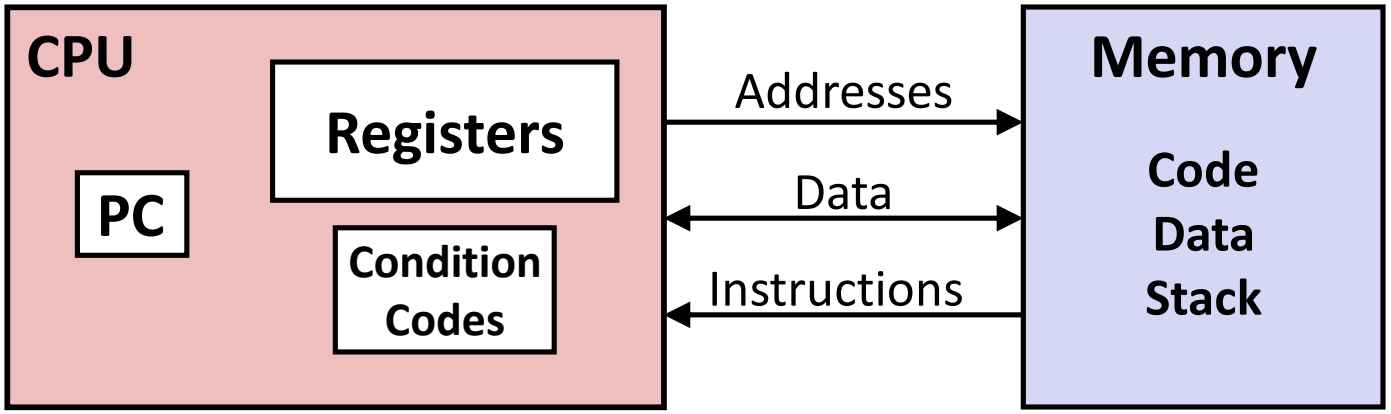
Today: Machine Programming I: Basics

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Definitions

- **Architecture: (also ISA: instruction set architecture) The parts of a processor design that one needs to understand or write assembly/machine code.**
  - Examples: instruction set specification, registers.
- **Microarchitecture: Implementation of the architecture.**
  - Examples: cache sizes and core frequency.
- **Code Forms:**
  - Machine Code: The byte-level programs that a processor executes
  - Assembly Code: A text representation of machine code
- **Example ISAs:**
  - Intel: x86, IA32, Itanium, x86-64
  - ARM: Used in almost all mobile phones
  - RISC V: New open-source ISA

Assembly/Machine Code View



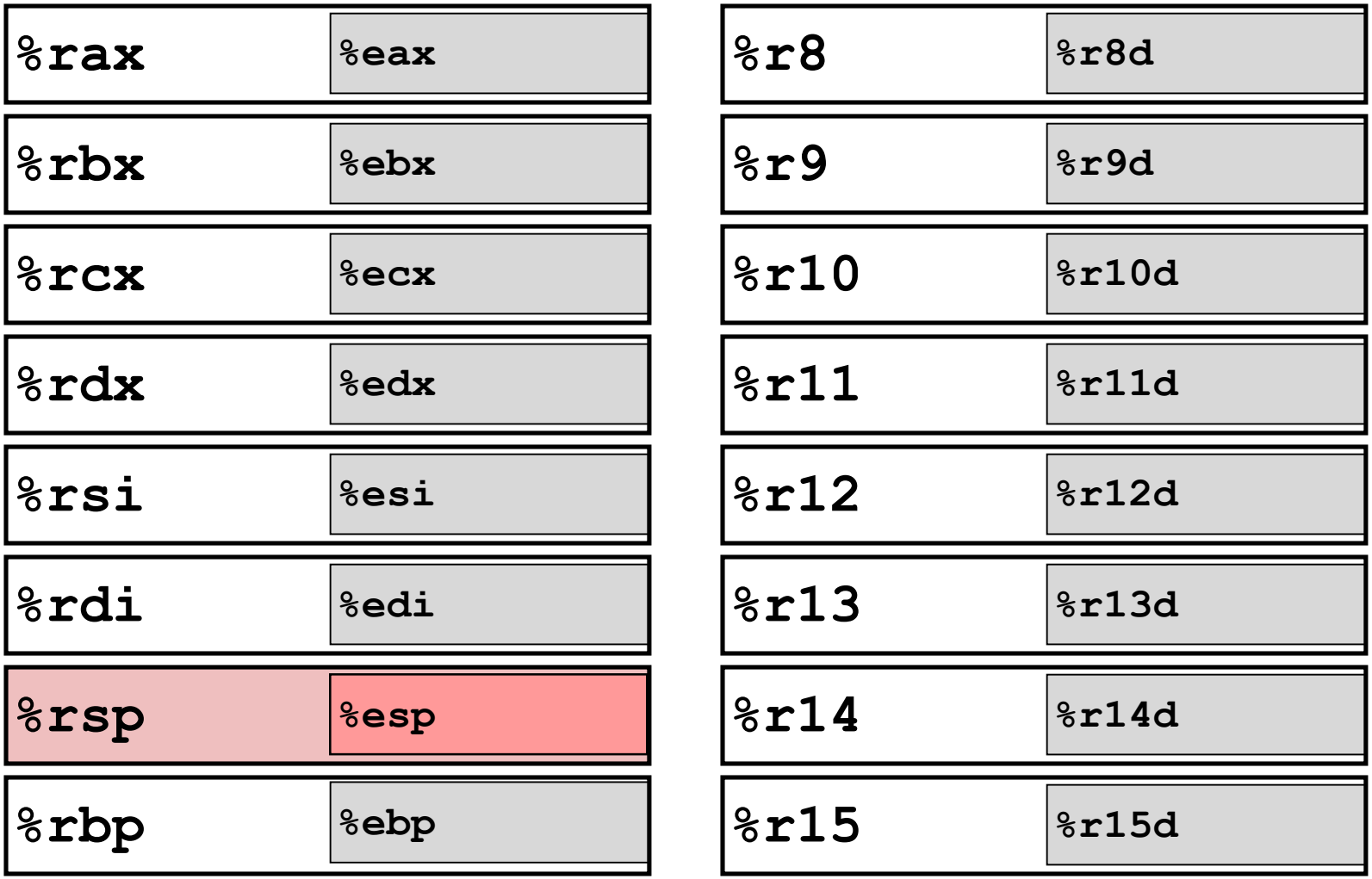
Programmer-Visible State

- **PC: Program counter**
  - Address of next instruction
  - Called "RIP" (x86-64)
- **Register file**
  - Heavily used program data
- **Condition codes**
  - Store status information about most recent arithmetic or logical operation
  - Used for conditional branching
- **Memory**
  - Byte addressable array
  - Code and user data
  - Stack to support procedures

Assembly Characteristics: Data Types

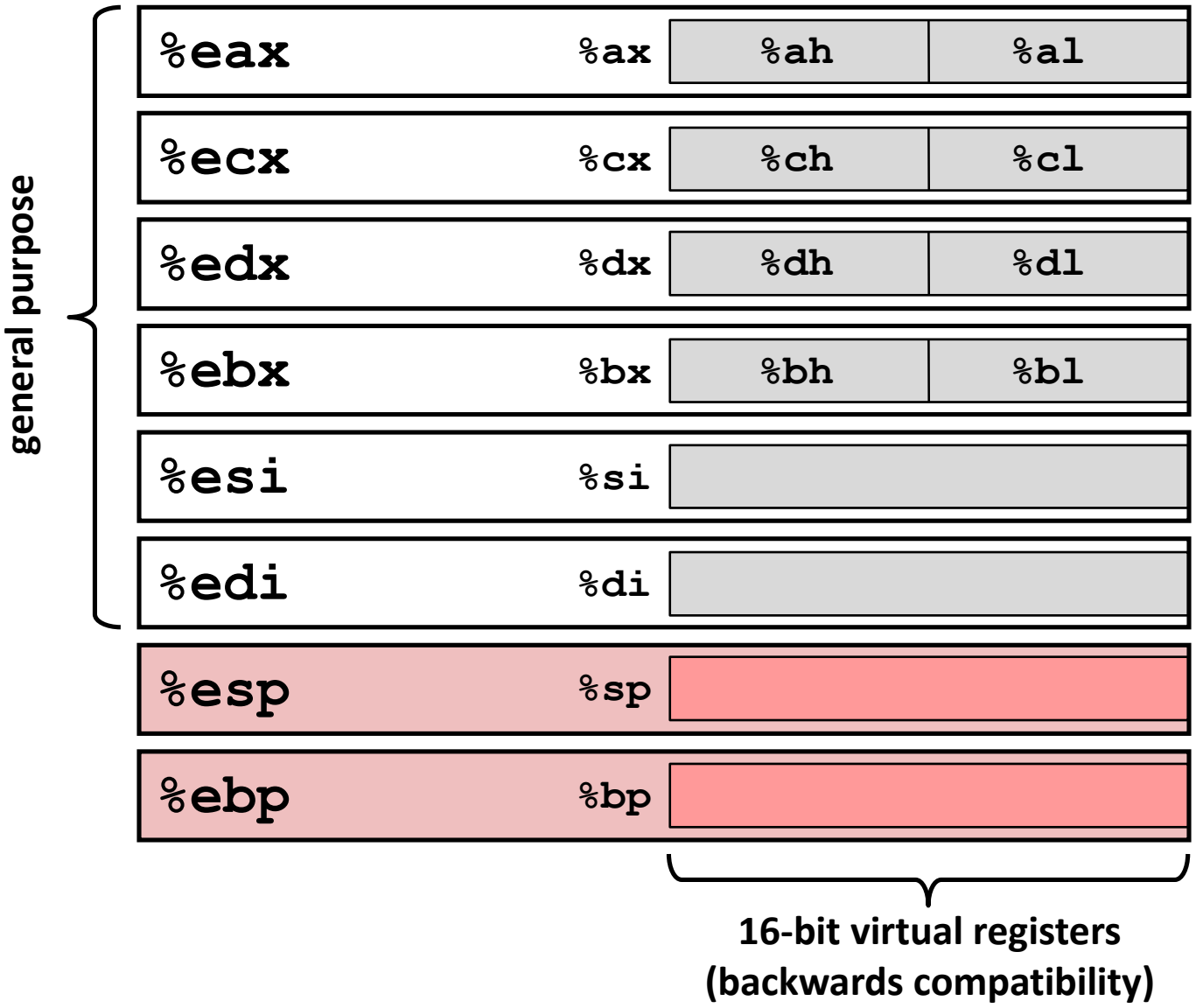
- **“Integer” data of 1, 2, 4, or 8 bytes**
  - Data values
  - Addresses (untyped pointers)
- **Floating point data of 4, 8, or 10 bytes**
- **Code: Byte sequences encoding series of instructions**
- **No aggregate types such as arrays or structures**
  - Just contiguously allocated bytes in memory

x86-64 Integer Registers



- Can reference low-order 4 bytes (also low-order 1 & 2 bytes)
- Not part of memory (or cache)

Some History: IA32 Registers



Assembly Characteristics: Operations

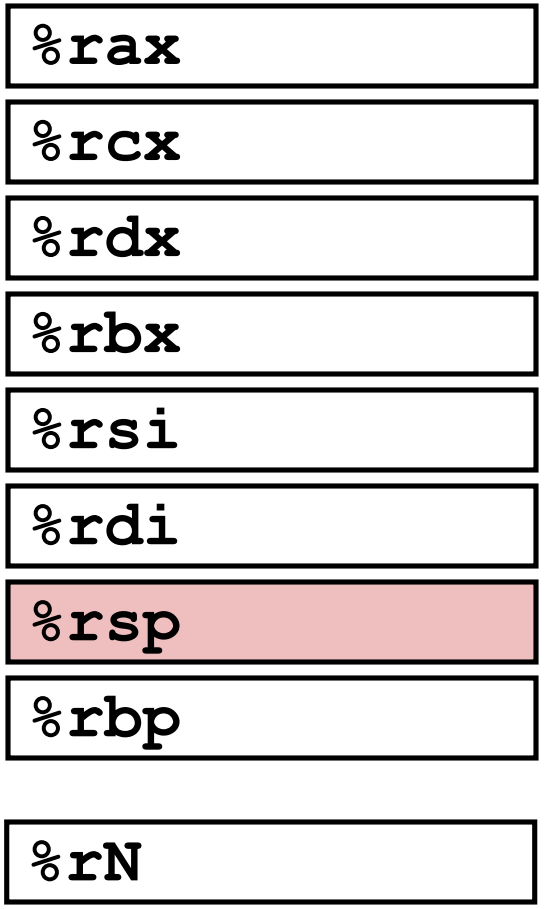
- **Transfer data between memory and register**
  - Load data from memory into register
  - Store data from register into memory
- **Perform arithmetic function on register or memory data**
- **Transfer control**
  - Unconditional jumps to/from procedures
  - Conditional branches
  - Indirect branches

Moving Data

- **Moving Data**

  `movq` ***Source*, *Dest*:**

- **Operand Types**
  - ***Immediate:*** Constant integer data
    - Example: `$0x400`, `$-533`
    - Like C constant, but prefixed with `'$'`
    - Encoded with 1, 2, or 4 bytes
  - ***Register:*** One of 16 integer registers
    - Example: `%rax`, `%r13`
    - But `%rsp` reserved for special use
    - Others have special uses for particular instructions
  - ***Memory:*** 8 consecutive bytes of memory at address given by register
    - Simplest example: `(%rax)`
    - Various other "address modes"



Warning: Intel docs use mov *Dest, Source*

movq Operand Combinations

	Source	Dest	Src,Dest	C Analog
movq	*Imm*	*Reg*	`movq $0x4,%rax`	`temp = 0x4;`
		Mem	`movq $-147,(%rax)`	`*p = -147;`
	Reg	*Reg*	`movq %rax,%rdx`	`temp2 = temp1;`
		Mem	`movq %rax,(%rdx)`	`*p = temp;`
	Mem	*Reg*	`movq (%rax),%rdx`	`temp = *p;`

Cannot do memory-memory transfer with a single instruction

Simple Memory Addressing Modes

- **Normal (R) Mem[Reg[R]]**
  - Register R specifies memory address
  - Pointer dereferencing in C

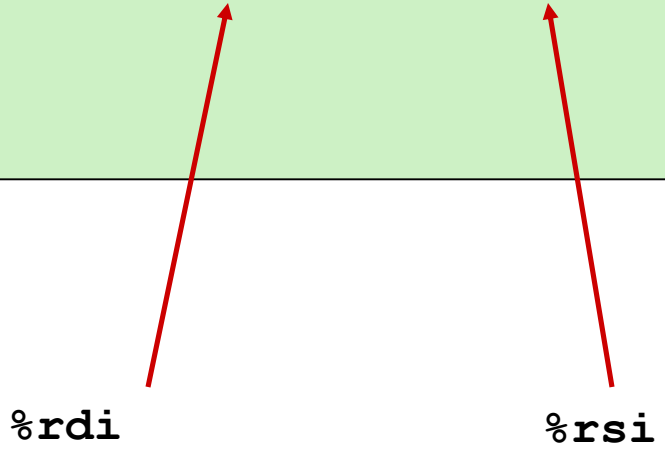
```
movq (%rcx),%rax
```

- **Displacement D(R) Mem[Reg[R]+D]**
  - Register R specifies start of memory region
  - Constant displacement D specifies offset

```
movq 8(%rbp),%rdx
```

Example of Simple Addressing Modes

```
void aFunc(<type> a, <type> b){
     ????
}
```



Example of Simple Addressing Modes

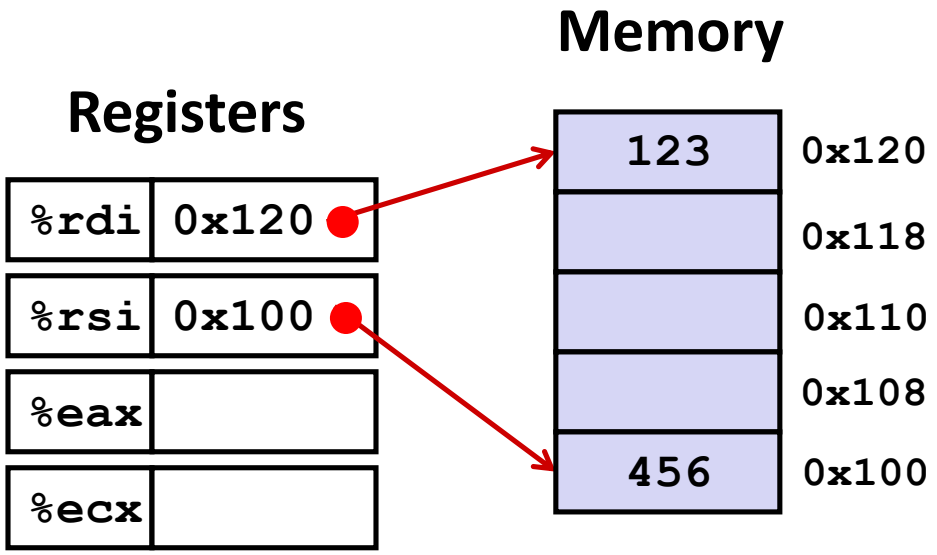
```
void swap (int *xp, int *yp){
  int t0 = *xp;
  int t1 = *yp;
  *xp = t1;
  *yp = t0;
}
```

```
swap:
   movl    (%rdi), %eax
   movl    (%rsi), %ecx
   movl    %ecx, (%rdi)
   movl    %eax, (%rsi)
   ret
```

Understanding Swap()

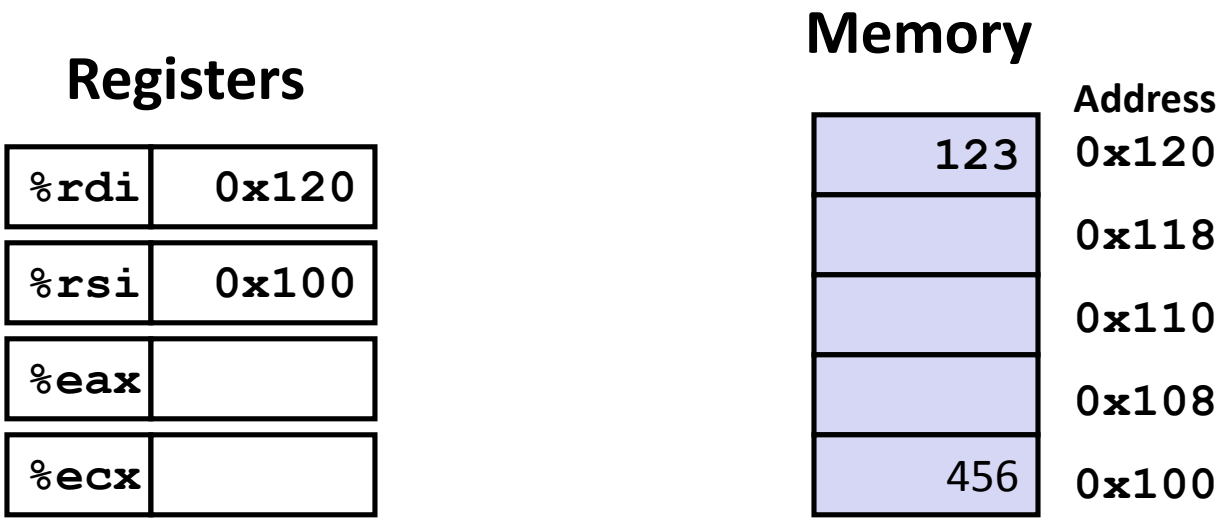
```
void swap (int *xp, int *yp){
   int t0 = *xp;
   int t1 = *yp;
   *xp = t1;
   *yp = t0;
}
```

Register	Value
%rdi	xp
%rsi	yp
%eax	t0
%ecx	t1



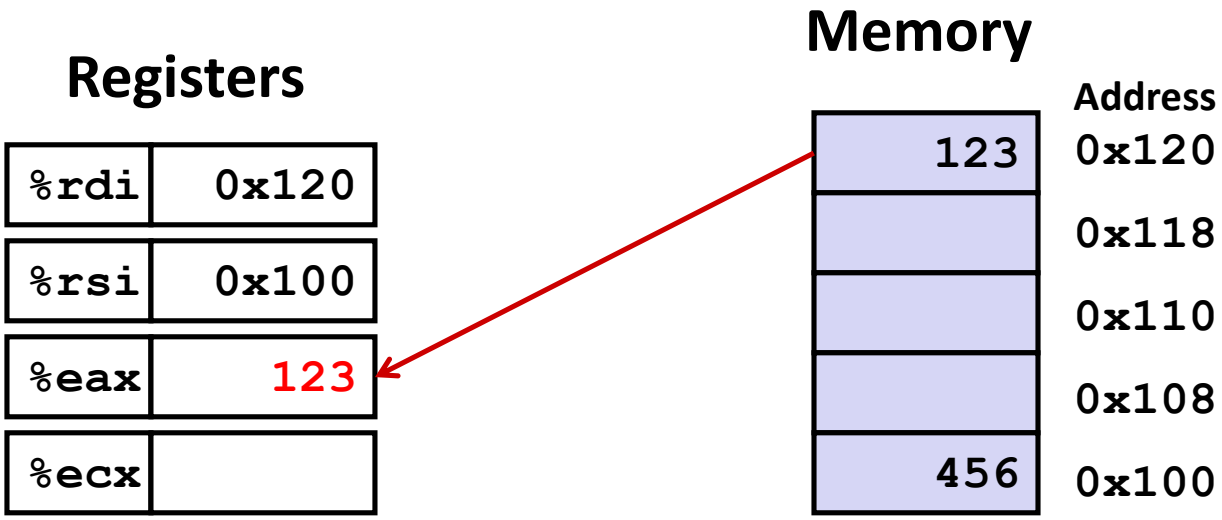

```
swap:
   movl    (%rdi), %eax      # t0 = *xp
   movl    (%rsi), %ecx      # t1 = *yp
   movl    %ecx, (%rdi)      # *xp = t1
   movl    %eax, (%rsi)      # *yp = t0
   ret
```

Understanding Swap()



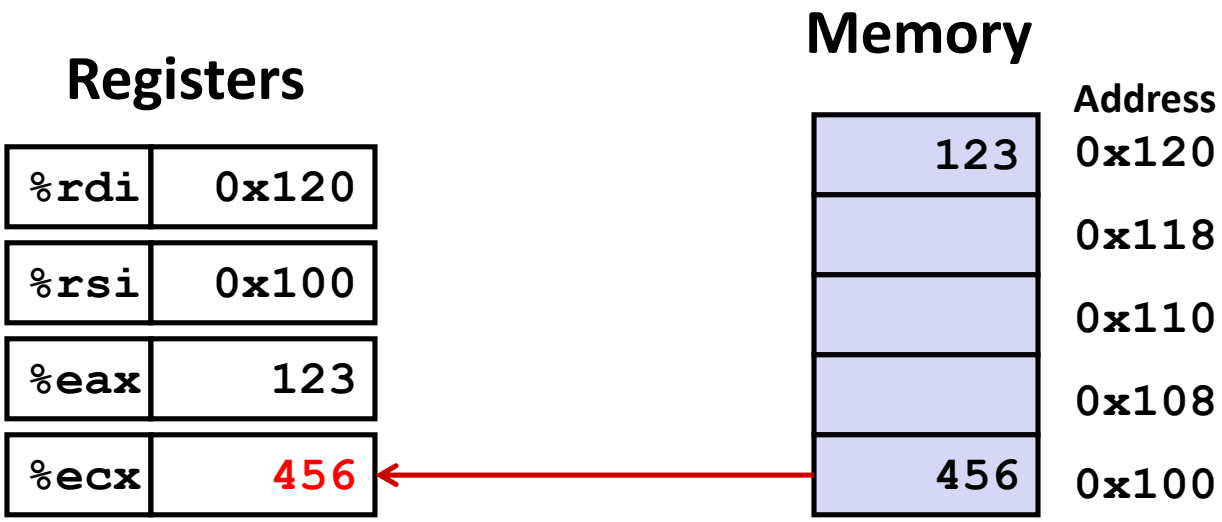

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   movl    (%rdi), %eax      # t0 = *xp
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   ret
```

Understanding Swap()



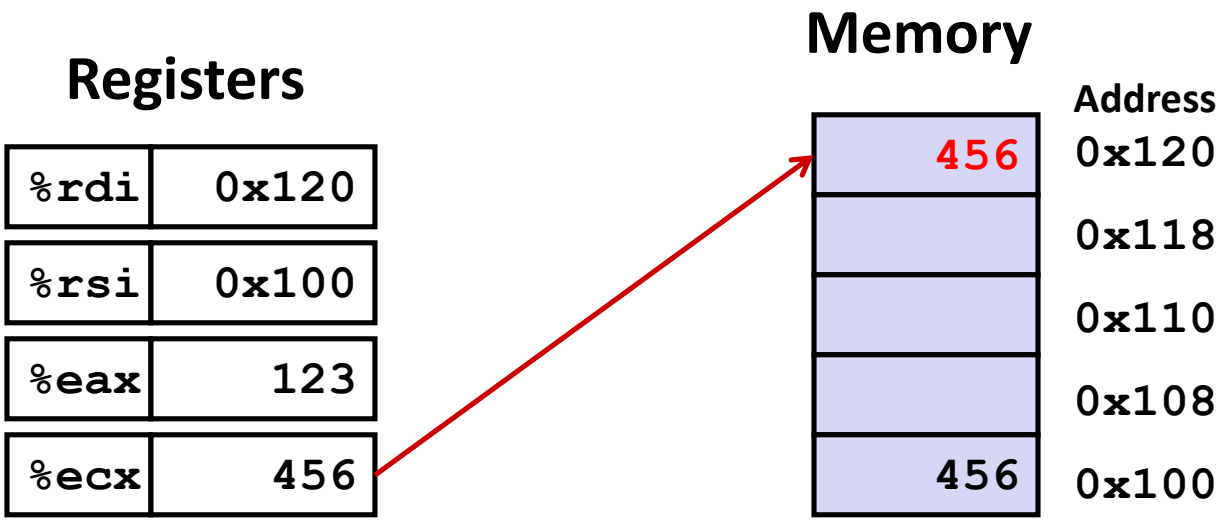

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   movl    (%rdi), %eax      # t0 = *xp
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Understanding Swap()



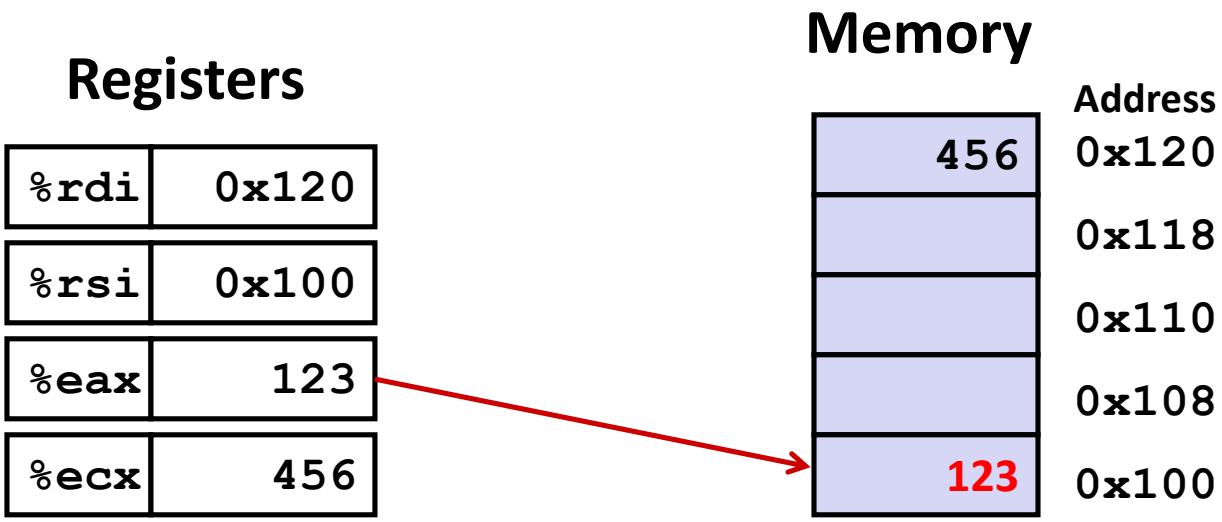

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   movl    (%rdi), %eax      # t0 = *xp
   movl    (%rsi), %ecx      # t1 = *yp
   movl    %ecx, (%rdi)      # *xp = t1
   movl    %eax, (%rsi)      # *yp = t0
   ret
```

Understanding Swap()




```
swap:
   movl    (%rdi), %eax      # t0 = *xp
   movl    (%rsi), %ecx      # t1 = *yp
   movl    %ecx, (%rdi)      # *xp = t1
   movl    %eax, (%rsi)      # *yp = t0
   ret
```

Understanding Swap()




```
swap:
   movl    (%rdi), %eax      # t0 = *xp
   movl    (%rsi), %ecx      # t1 = *yp
   movl    %ecx, (%rdi)      # *xp = t1
   movl    %eax, (%rsi)      # *yp = t0
   ret
```

Complete Memory Addressing Modes

- **Most General Form**

  D(Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]+ D]**

  - D: Constant "displacement" 1, 2, or 4 bytes
  - Rb: Base register: Any of 16 integer registers
  - Ri: Index register: Any, except for `%rsp`
  - S: Scale: 1, 2, 4, or 8 (*why these numbers?*)

- **Special Cases**

  (Rb,Ri) **Mem[Reg[Rb]+Reg[Ri]]**

  D(Rb,Ri) **Mem[Reg[Rb]+Reg[Ri]+D]**

  (Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]]**

Address Computation Examples

%rdx	0xf000
%rcx	0x0100

D(Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]+ D]**

- D: Constant "displacement" 1, 2, or 4 bytes
- Rb: Base register: Any of 16 integer registers
- Ri: Index register: Any, except for %rsp
- S: Scale: 1, 2, 4, or 8 (*why these numbers?*)

Expression	Address Computation	Address
0x8(%rdx)	0xf000 + 0x8	0xf008
(%rdx,%rcx)	0xf000 + 0x100	0xf100
(%rdx,%rcx,4)	0xf000 + 4*0x100	0xf400
0x80(,%rdx,2)	2*0xf000 + 0x80	0x1e080

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Address Computation Instruction

- **`leaq` *Src*, *Dst***
  - *Src* is address mode expression
  - Set *Dst* to address denoted by expression

- **Uses**
  - Computing addresses without a memory reference
    - E.g., translation of `p = &x[i];`
  - Computing arithmetic expressions of the form **x + k*y**
    - k = 1, 2, 4, or 8

- **Example**

```
long m12(long x){
  return x*12;
}
```

Converted to ASM by compiler:

```
leaq (%rdi,%rdi,2), %rax # t <- x+x*2
salq $2, %rax            # return t<<2
```

Some Arithmetic Operations

- **Two Operand Instructions:**

Format		*Computation*	
`addq`	`Src,Dest`	Dest = Dest + Src	
`subq`	`Src,Dest`	Dest = Dest – Src	
`imulq`	`Src,Dest`	Dest = Dest * Src	
`salq`	`Src,Dest`	Dest = Dest << Src	*Also called shlq*
`sarq`	`Src,Dest`	Dest = Dest >> Src	*Arithmetic right shift*
`shrq`	`Src,Dest`	Dest = Dest >> Src	*Logical right shift*
`xorq`	`Src,Dest`	Dest = Dest ^ Src	
`andq`	`Src,Dest`	Dest = Dest & Src	
`orq`	`Src,Dest`	Dest = Dest | Src	

- **Watch out for argument order!**

  (Warning: Intel docs use "op Dest,Src")

- **No distinction between signed and unsigned int (why?)**

Some Arithmetic Operations

- **One Operand Instructions**

`incq`	`Dest`	Dest = Dest + 1
`decq`	`Dest`	Dest = Dest – 1
`negq`	`Dest`	Dest = – Dest
`notq`	`Dest`	Dest = ~Dest

- **See book for more instructions**

Arithmetic Expression Example

```
long arith
(long x, long y, long z){
  long t1 = x+y;
  long t2 = z+t1;
  long t3 = x+4;
  long t4 = y * 48;
  long t5 = t3 + t4;
  long rval = t2 * t5;
  return rval;
}
```

```
arith:
  leaq    (%rdi,%rsi), %rax
  addq    %rdx, %rax
  leaq    (%rsi,%rsi,2), %rdx
  salq    $4, %rdx
  leaq    4(%rdi,%rdx), %rcx
  imulq   %rcx, %rax
  ret
```

Interesting Instructions

- **`leaq`**: address computation
- **`salq`**: shift
- **`imulq`**: multiplication
  - But, only used once

Understanding Arithmetic Expression Example

```
long arith
(long x, long y, long z){
  long t1 = x+y;
  long t2 = z+t1;
  long t3 = x+4;
  long t4 = y * 48;
  long t5 = t3 + t4;
  long rval = t2 * t5;
  return rval;
}
```

```
arith:
  leaq    (%rdi,%rsi), %rax    # t1
  addq    %rdx, %rax           # t2
  leaq    (%rsi,%rsi,2), %rdx
  salq    $4, %rdx             # t4
  leaq    4(%rdi,%rdx), %rcx   # t5
  imulq   %rcx, %rax           # rval
  ret
```

Register	Use(s)
%rdi	Argument x
%rsi	Argument y
%rdx	Argument z, t4
%rax	t1, t2, rval
%rcx	t5

Compiler optimization:

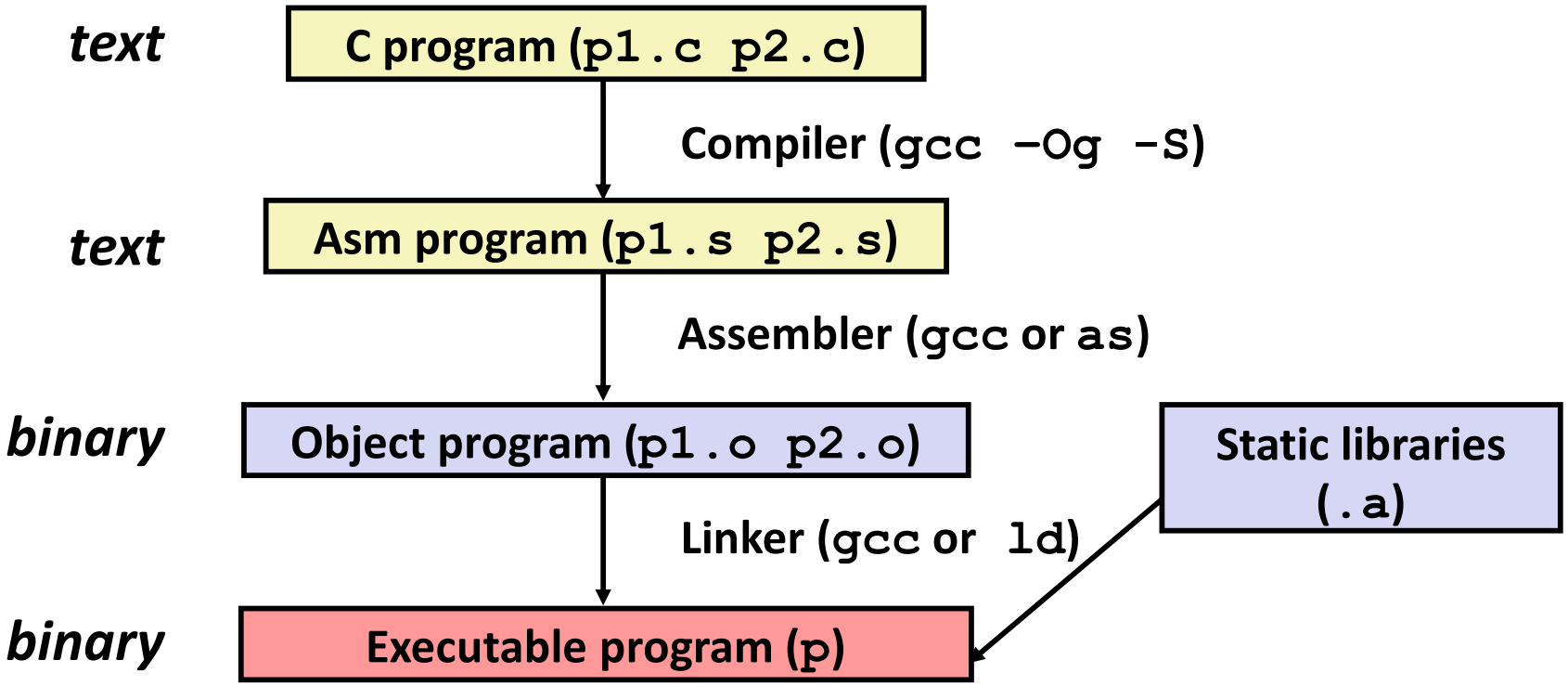
- Reuse of registers
- Substitution (copy propagation)
- Strength reduction

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Turning C into Object Code

- Code in files `p1.c p2.c`
- Compile with command: `gcc –Og p1.c p2.c -o p`
  - Use basic optimizations (`–Og`) [New to recent versions of GCC]
  - Put resulting binary in file `p`



Compiling Into Assembly

C Code (sum.c)

```
int plus(int x, int y){
    return x+y;
}
void sumstore(int x, int y, int *dest){
    int t = plus(x, y);
    *dest = t;
}
```

Generated x86-64 Assembly

```
sumstore:
    pushq   %rbp
    movq    %rsp, %rbp
    addl    %esi, %edi
    movl    %edi, (%rdx)
    popq    %rbp
    ret
```

Obtain (on Mac OS) with command

```
gcc -O -S sum.c
```

Produces file `sum.s`

Warning: **Will get very different results on different machines (Linux, Mac OS-X, …) due to different versions of gcc and different compiler settings.**

What it Really Looks Like

```
        .globl  _sumstore
        .align  4, 0x90
_sumstore:
        .cfi_startproc
## BB#0:
        pushq   %rbp
Ltmp3:
        .cfi_def_cfa_offset 16
Ltmp4:
        .cfi_offset %rbp, -16
        movq    %rsp, %rbp
Ltmp5:
        .cfi_def_cfa_register %rbp
        addl    %esi, %edi
        movl    %edi, (%rdx)
        popq    %rbp
        retq
        .cfi_endproc
```

Things preceding with a "." are generally directives

```
sumstore:
    pushq   %rbp
    movq    %rsp, %rbp
    addl    %esi, %edi
    movl    %edi, (%rdx)
    popq    %rbp
    ret
```

Object Code

Code for `sumstore`

```
0x100000f20:
    0x55
    0x48
    0x89
    0xe5
    0x01
    0xf7
    0x89
    0x3a
    0x5d
    0xc3
```

- **Total of 10 bytes**
- **Each instruction 1, 2, or 3 bytes**
- **Starts at address `0x100000f20`**

- **Assembler**
  - Translates `.s` into `.o`
  - Binary encoding of each instruction
  - Nearly-complete image of executable code
  - Missing linkages between code in different files
- **Linker**
  - Resolves references between files
  - Combines with static run-time libraries
    - E.g., code for **`malloc`**, **`printf`**
  - Some libraries are *dynamically linked*
    - Linking occurs when program begins execution

Machine Instruction Example

```
*dest = t;
```

- **C Code**
  - Store value **t** where designated by **dest**

```
movl    %edi, (%rdx)
```

- **Assembly**
  - Move 4-byte value to memory
  - Operands:
    - **t:** Register **%edi**
    - **dest:** Register **%rdx**
    - ***dest:** Memory **M[%rdx]**

```
0x100000f26: 89 3a
```

- **Object Code**
  - 2-byte instruction
  - Stored at address **0x100000f26**

Disassembling Object Code

- **Disassembler**

  `objdump -d sum`

  - Useful tool for examining object code
  - Analyzes bit pattern of series of instructions
  - Produces approximate rendition of assembly code
  - Can be run on either `a.out` (complete executable) or `.o` file

Disassembled

```
<sumstore>:
  100000f20:  55              pushq  %rbp
  100000f21:  48 89 e5        movq   %rsp, %rbp
  100000f24:  01 f7           addl   %esi, %edi
  100000f26:  89 3a           movl   %edi, (%rdx)
  100000f28:  5d              popq   %rbp
  100000f29:  c3              retq
```

Alternate Disassembly

Object

```
0x100000f20:
    0x55
    0x48
    0x89
    0xe5
    0x01
    0xf7
    0x89
    0x3a
    0x5d
    0xc3
```

Disassembled

```
Dump of assembler code for function sumstore:
 0x0000000100000f20 <+0>: pushq   %rbp
 0x0000000100000f21 <+1>: movq    %rsp,%rbp
 0x0000000100000f24 <+4>: addl    %esi, %edi
 0x0000000100000f26 <+6>: movl    %edi, (%rdx)
 0x0000000100000f28 <+8>: popq    %rbp
 0x0000000100000f29 <+9>: retq
```

- **Within gdb Debugger**
  - **`gdb sum`**
  - **`disassemble sumstore`**
    - Disassemble procedure
  - **`x/10xb sumstore`**
    - Examine the 10 bytes starting at `sumstore`

What Can be Disassembled?

```
% objdump -d WINWORD.EXE

WINWORD.EXE:    file format pei-i386

No symbols in "WINWORD.EXE".
Disassembly of section .text:

30001000 <.text>:
30001000:
30001001:
30001003:
30001005:
3000100a:
```

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- Anything that can be interpreted as executable code
- Disassembler examines bytes and reconstructs assembly source

Machine Programming I: Summary

- **History of Intel processors and architectures**
  - Evolutionary design leads to many quirks and artifacts
- **C, assembly, machine code**
  - New forms of visible state: program counter, registers, ...
  - Compiler must transform statements, expressions, procedures into low-level instruction sequences
- **Assembly Basics: Registers, operands, move**
  - The x86-64 move instructions cover wide range of data movement forms
- **Arithmetic**
  - C compiler will figure out different instruction combinations to carry out computation